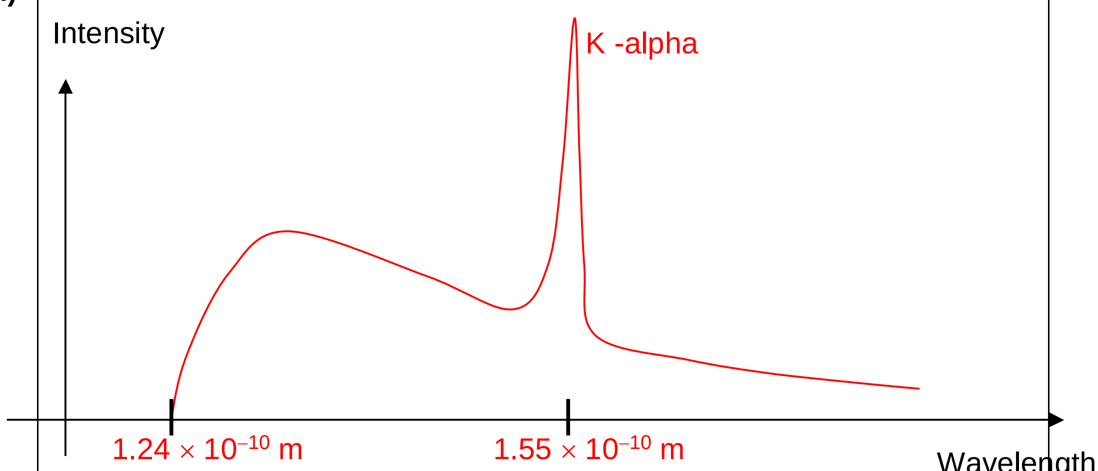
9	**(a)**	**(i)**	1. Using $e\,V_{acc} = h\,c\,/\,\lambda_{min}$, [1] we have $\lambda_{min} = (6.63 \times 10^{-34})\,(3.0 \times 10^{8})\,/\,(1.60 \times 10^{-19} \times 10 \times 10^{3})$ [1] $= 1.24 \times 10^{-10}$ m [1]	
		(i)	2. Using $\Delta E_{(2\rightarrow 1)} = h\,c\,/\,\lambda$, we have $\lambda = (6.63 \times 10^{-34})\,(3.0 \times 10^{8})\,/\,[\,(-933) - (-8979)\,] \times 1.60 \times 10^{-19}$ [1] $= 1.55 \times 10^{-10}$ m [1]	
	(a)	**(ii)**	 Shape [1] Minimum wavelength labelled [1] K-alpha wavelength labelled [1]	
	(b)	Using formula $\Delta p_x\,\Delta x \geq h$, $\Delta p_x \geq 6.63 \times 10^{-34} / (1 \times 10^{-15}) = 6.63 \times 10^{-19}$ [1] $\Delta v \geq 6.63 \times 10^{-19} / (9.11 \times 10^{-31}) = 7.3 \times 10^{11}$ m s^{-1} [1] Thus, $\Delta v > c$, which is impossible [1]		
	(c)	${}^{45}_{20}Ca \rightarrow {}^{45}_{21}Sc + {}^{0}_{-1}e$ [1] (Negative) beta decay [1]		

	(d)	Mass difference = 219.009523 – (214.999469 + 4.002603) [1] = 0.007451 u Energy released = $(0.007451 \times 1.66 \times 10^{-27})(3 \times 10^{8})^{2}$ [1] = 1.1132×10^{-12} J = 6.96 MeV [1]	
	(e)	Without a second emitted particle, all β-particles would have the same energy [1] Range of β-particle energy values means range of momenta (speeds) [1] For COM and COE [1], there must be another particle to share the momentum and energy [1]	